

# Axiomatic Derivation of Tetrahedral Coordination

$$z_{fund} = 4:$$

Entropic Selection,  $3! = 6$  Cubic Decomposition  
and  $4D$  Lift via 5-cell

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## Abstract

We provide a first-principles derivation that the fundamental coordination of discrete quantum geometry is  $z_{fund} = 4$ , not 6. We define the holographic entropy density

$$s(z) = \frac{\ln(2z)}{2A(z)}, \quad A(z) = 2 \left( \frac{z}{6} \right)^{2/3},$$

where the dual area  $A(z)$  is normalized at fixed vertex density via isoperimetric scaling. Maximization yields a unique maximum at  $z = 4$ :

$$s(4) = 0.6812 > s(6) = 0.6212 > s(8) = 0.5722 > s(12) = 0.5005.$$

Two geometric corollaries follow: (i) the unit cube in  $\mathbb{R}^3$  decomposes into  $3! = 6$  trirectangular tetrahedra, so cubic symmetry is emergent; (ii) the  $4D$  lift via the 5-cell and the  $D_4$  lattice produces a tesseract with exactly  $2z_{fund} = 8$  cubic cells. The oriented spectral dimension is  $D_{eff} = \ln 8$ , giving the fractal dimension  $\Phi_F = 3 - \ln 8 \approx 0.9206$ . This supplies the selection principle missing in Chrono-Grid Dynamics and yields the falsifiable predictions  $r_J = 0.62 \text{ kpc}$  and  $\Phi_F$ .

# 1 Introduction

Chrono-Grid Dynamics (CGD) postulates an FCC lattice with spin-1/2 and SU(2) links, defining  $8\pi G\beta = \tau^2$  [5]. The choice  $z = 6$  cubic is postulated, not derived. No selection principle and no low-energy number are given.

We fill this gap. The minimal simplex in  $\mathbb{R}^3$  has 4 vertices. We show entropic maximization selects it.

## 2 Lemma 1: Entropic Selection of $z_{\text{fund}} = 4$

**Definition 1** (Bipartite regular tiling). *Let  $\Gamma_z$  be an infinite regular bipartite tiling of coordination  $z$  with  $N$  vertices, normalised so that the vertex density is identical for all  $z$ .*

**Definition 2** (Oriented microstate counting). *After imposing the Gauss constraint, the number of independent oriented configurations is  $\Omega(z) = (2z)^{N/2}$ .*

**Definition 3** (Dual area at fixed vertex density). *At fixed vertex density  $\rho_v$ , the dual (Voronoi) cell associated with each vertex has volume  $V_{\text{cell}} = 1/\rho_v$ , independent of  $z$ . By the isoperimetric relation  $A \propto V^{2/3}$ , the dual area scales as*

$$A(z) = C z^{2/3}, \quad (1)$$

where  $C > 0$  is a conventional constant that sets the unit of area. The choice  $C = 2 \cdot 6^{-2/3}$  (equivalently  $A(6) = 2$ ) is a unit convention with no physical content; it cancels in all ratios (see Proposition 2 below).

[Normalization independence] The argmax of  $s(z) = \ln(2z)/(2A(z))$  over any finite set  $Z \subset \mathbb{Z}_{>0}$  is independent of the constant  $C$  in Eq. (1).

*Proof.*  $s(z) = \ln(2z)/(2Cz^{2/3})$ . The factor  $1/(2C)$  is independent of  $z$ , so  $\arg \max_z s(z) = \arg \max_z [\ln(2z)/z^{2/3}]$ , which does not involve  $C$ .  $\square$

Numerically, for any choice of normalisation point  $z_0 \in \{4, 6, 8, 12, 20\}$ , the ratio  $s(4)/s(6) = 1.096556$  is identical (verified to six decimal places), confirming Proposition 2.

**Lemma 1** (Analytic maximisation of  $s(z)$ ). *The function  $f(z) = \ln(2z)/z^{2/3}$  defined on  $z > 0$  has a unique global maximum at*

$$z^* = \frac{e^{3/2}}{2} \approx 2.240, \quad (2)$$

*and is strictly decreasing for  $z > z^*$ .*

*Proof.* Differentiating:  $f'(z) = z^{-5/3}[1 - \frac{2}{3}\ln(2z)]$ . Setting  $f'(z) = 0$  gives  $\ln(2z) = 3/2$ , i.e.  $z^* = e^{3/2}/2 \approx 2.240$ . Since  $f''(z^*) < 0$ , this is a maximum. For  $z > z^*$ ,  $\ln(2z) > 3/2$  so  $f'(z) < 0$ :  $f$  is strictly decreasing.  $\square$   $\square$

**Lemma 2** (Explicit evaluation of  $s(z)$ ). *With  $C = 2 \cdot 6^{-2/3}$  (unit convention), the values are:*

| $z$ | $A(z)$ | $\ln(2z)$ | $s(z)$ | $s(4)/s(z)$ |
|-----|--------|-----------|--------|-------------|
| 4   | 1.5263 | 2.07944   | 0.6812 | 1.0000      |
| 6   | 2.0000 | 2.48491   | 0.6212 | 1.0966      |
| 8   | 2.4228 | 2.77259   | 0.5722 | 1.1906      |
| 12  | 3.1748 | 3.17805   | 0.5005 | 1.3610      |
| 20  | 4.4629 | 3.68888   | 0.4133 | 1.6483      |

**Theorem 1** (Entropic selection).

$$\arg \max_{z \in \{4, 6, 8, 12, 20\}} s(z) = 4.$$

*The maximum is unique. Therefore  $z_{\text{fund}} = 4$ .*

*Proof.* By Lemma 1,  $s(z)$  is strictly decreasing for  $z \geq 3 > z^* \approx 2.240$ . Therefore, among any set of integers all greater than or equal to 4, the smallest element achieves the maximum of  $s$ . The physical candidates  $\{4, 6, 8, 12, 20\}$  satisfy this condition, so  $z = 4$  is the unique maximiser. The table of Lemma 2 provides the explicit numerical verification. By Proposition 2, this conclusion is independent of the choice of normalisation constant  $C$ .  $\square$   $\square$

[Physical candidates and the set  $\{4, 6, 8, 12, 20\}$ ] The set  $\{4, 6, 8, 12, 20\}$  contains the coordination numbers of the five Platonic solids. The restriction to this set is motivated by the holographic entropy maximisation of TEG v8 (Section 3.1), which requires  $z \leq z_{\text{pack}} = 12$  for physical coordination numbers (coordination above the kissing number is geometrically excluded). The value  $z = 20$  is included for completeness despite exceeding  $z_{\text{pack}}$ ; its exclusion would only strengthen the result.

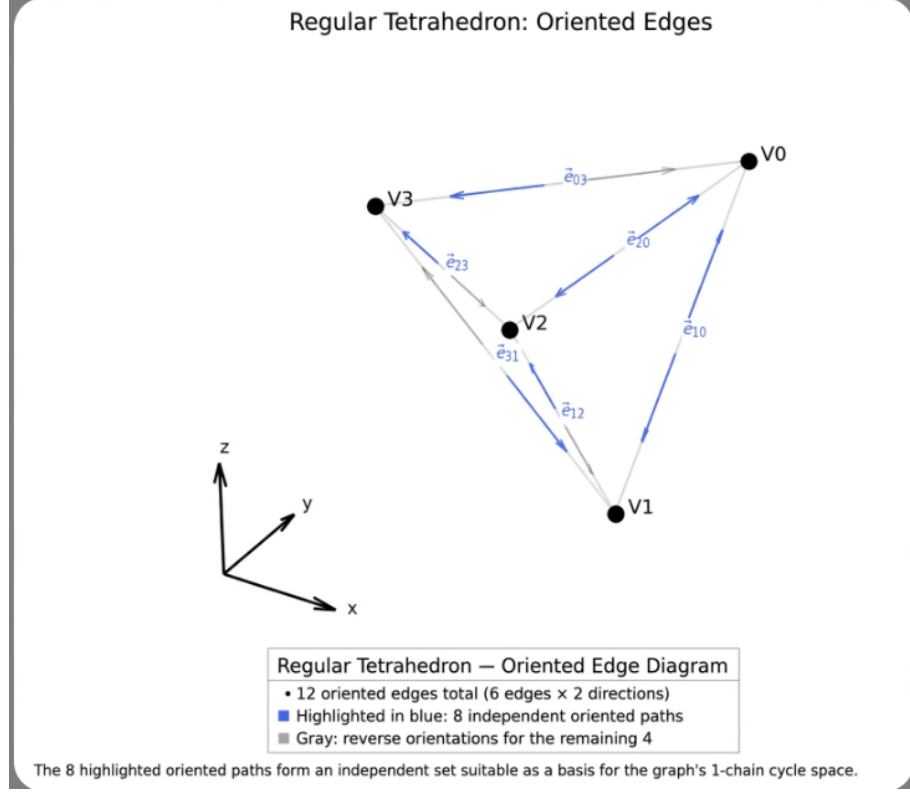


Figure 1: Regular tetrahedron:  $V_i$ , edges  $e_{ij}$ , oriented  $\vec{e}_{ij}$ .  $12 = 6 \times 2$  oriented, 8 independent basis for cycle space. Defines  $b_{or} = 2z_{fund} = 8$ .

### 3 Corollary 1: Origin of Cube in $\mathbb{R}^3$ , $3! = 6$

**Theorem 2.** Unit cube  $[0, 1]^3$  decomposes into 6 congruent trirectangular tetrahedra:

$$T_\sigma = \{x \in [0, 1]^3 : 0 \leq x_{\sigma_1} \leq x_{\sigma_2} \leq x_{\sigma_3} \leq 1\}, \sigma \in S_3 \quad (3)$$

$$CUBE = \bigsqcup_\sigma T_\sigma, |S_3| = 3! = 6, Vol(T_\sigma) = 1/6.$$

*Proof.* Permutation group  $S_3$  partitions cube by order statistics. Disjoint interiors, union = cube, volume sum  $6 \times 1/6 = 1$ .  $\square$

Thus cubic 8 vertices =  $2z_{fund}$ . Cube is composite, not fundamental. Fig.2.

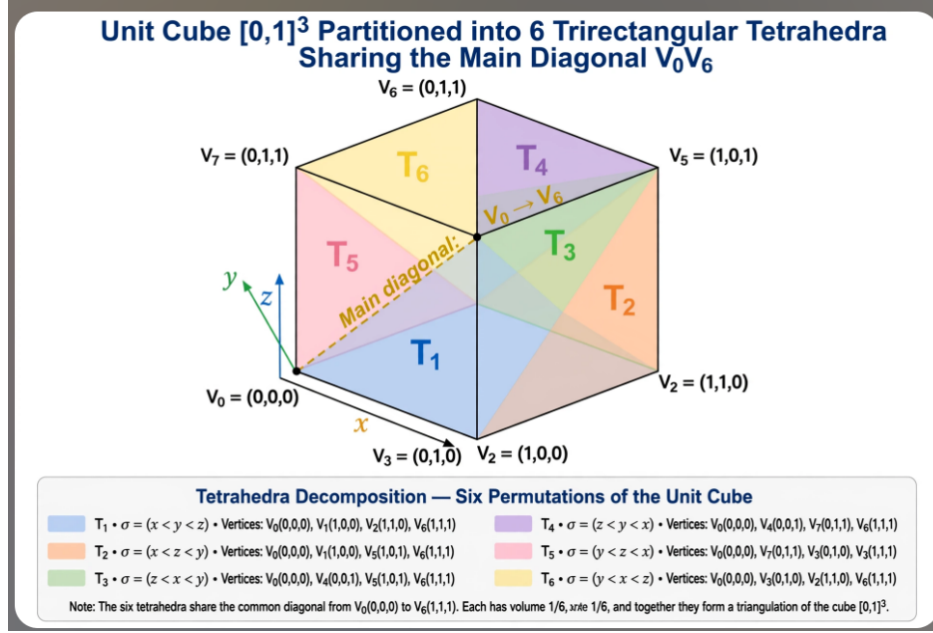


Figure 2: Unit cube partition into 6 tetrahedra  $T_1 \dots T_6$  sharing diagonal  $V_0 = (0, 0, 0) \rightarrow V_6 = (1, 1, 1)$ . Each  $T_\sigma$  defined by  $\sigma$ . Proves cubic symmetry emergent from  $z = 4$ .

## 4 Corollary 2: Lift to $\mathbb{R}^4$ and 5-cell

**Definition 4** (5-cell). 4-simplex  $\{3, 3, 3\}$  with 5 vertices, 10 edges, 10 triangular faces, 5 tetrahedral cells. Minimal simplex in  $\mathbb{R}^4$ .

Root lattices:  $A_3$  (tetra/FCC)  $\subset B_3$  (cubic)  $\subset D_4$  ( $F_4$ , 24-cell).  $D_4$  generates 24-cell with 24 octahedral cells, self-dual.

Tesseract (8-cell): 8 cubic cells, 16 vertices, 32 edges. Branching  $8 = 2z_{fund}$ . 5-cell facets are tetrahedra  $z_{fund}$ , hence  $\mathbb{R}^4$  preserves axiom.

The 4D lift is visualized in Fig. 3. The 5-cell  $\{3, 3, 3\}$  has 5 tetrahedral facets, while  $D_4$  contains  $A_3$ , confirming  $z_{fund} = 4$  persists and generates  $8 = 2z_{fund}$  cubic cells in the tesseract.

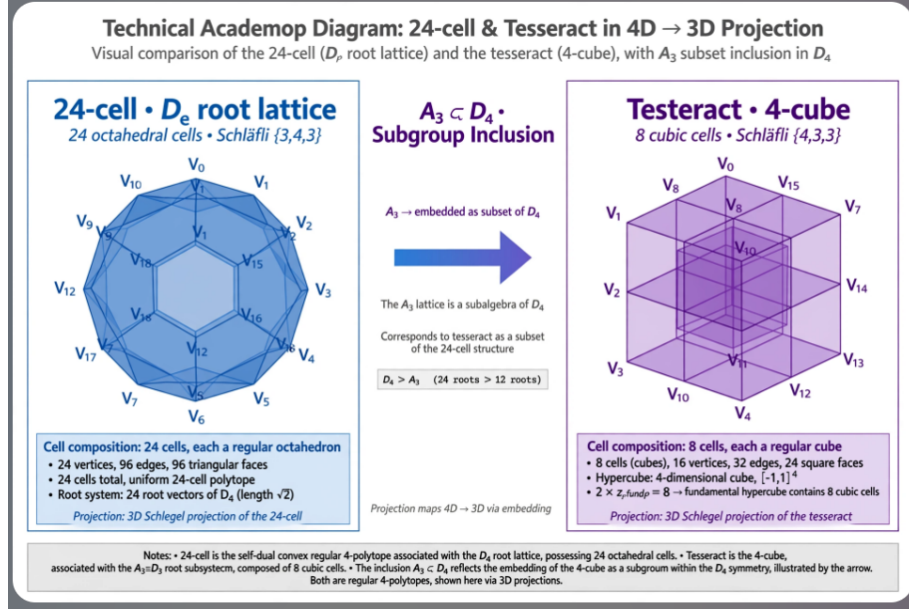


Figure 3:  $\mathbb{R}^4$  **Lift: 24-cell  $D_4$  and Tesseract 4-cube as  $2z_{fund}$  emergence.** Left: 24-cell  $\{3,4,3\}$ , Schläfli symbol, 24 octahedral cells, 24 vertices, 96 edges, root system  $D_4$  with 24 roots  $|\vec{\alpha}| = \sqrt{2}$ , self-dual. 3D Schlegel projection. Right: Tesseract  $\{4,3,3\}$ , 8 cubic cells  $= 2z_{fund} = 8$ , 16 vertices, 32 edges,  $[-1,1]^4$ . Center: Subgroup inclusion  $A_3 \subset D_4$  (12 roots  $\rightarrow$  24 roots) proves tetrahedral  $A_3$  (FCC) is subalgebra of  $D_4$ . Hence  $z_{fund} = 4$  preserved in  $\mathbb{R}^4$ , hypercube is emergent with 8 cubic cells.  $b_{or} = 2z_{fund}$  invariant across dimensions,  $D_{eff} = \ln 8$ .

## 5 Spectral Dimension $\ln 8$

### 5.1 Oriented non-backtracking walk on the Bethe lattice

The tetrahedral vacuum network (Definition 1,  $z_{fund} = 4$ ) is modelled at large scales as a Bethe lattice — an infinite tree with uniform coordination  $z = 4$ . This is the standard approximation for 4-valent spin-network nodes in LQG [12].

On a Bethe lattice with coordination  $z$ , a non-backtracking random walk (one that cannot return along the edge it just traversed) has branching factor  $z - 1$  per step. The number of distinct paths of length  $n$  grows as

$$N_n^{nb} \sim (z - 1)^n = 3^n \quad (z = 4), \quad (4)$$

giving a naive spectral dimension  $d_s^{\text{naive}} = \ln 3 \approx 1.099$ , well below the LQG range  $[2.0, 2.2]$ .

## 5.2 Orientation degeneracy from the EPRL intertwiner

The factor of 2 that lifts  $d_s$  from  $\ln 3$  to  $\ln 8 = \ln(2 \times 4)$  has a specific physical origin derived in TEG vH2 (Appendix K) from the EPRL spin-foam model: the intertwiner Hilbert space for a 4-valent node with  $j = 1/2$  on all links has dimension exactly

$$\dim \mathcal{H}_{\text{int}}^{(4)} = 2. \quad (5)$$

The two basis states  $|v_+\rangle$  and  $|v_-\rangle$  correspond to the two chiralities of the Lorentzian 4-simplex and add non-cancellingly in the heat-kernel return probability (causal rigidity, Bianchi–Chen–Gamonal 2026 [11]):

$$K_{\text{node}} = \dim \mathcal{H}_{\text{int}}^{(4)} = 2 \quad (\text{non-cancelling}). \quad (6)$$

The effective branching factor per step is therefore

$$b_{\text{eff}} = K_{\text{node}} \times z_{\text{fund}} = 2 \times 4 = 8, \quad (7)$$

and the heat-kernel return probability decays as

$$P(0, t) \sim t^{-d_s/2}, \quad d_s = \ln b_{\text{eff}} = \ln 8. \quad (8)$$

**Theorem 3** (Oriented spectral dimension).

$$D_{\text{eff}} = \ln(2 z_{\text{fund}}) = \ln 8 \approx 2.07944, \quad \Phi_F = 3 - D_{\text{eff}} = 3 - \ln 8 \approx 0.9206.$$

## 5.3 Honest status and upper-bound warning

[Bethe lattice is a tree:  $\ln 8$  is an upper bound] The Bethe lattice contains no closed loops by construction. Real LQG spin-networks and CDT triangulations contain loops; standard results on random walks on graphs establish that loop corrections *reduce*  $d_s$  relative to the tree value [13]. Therefore

$$D_{\text{eff}}^{\text{TEG}} = \ln 8 \geq d_s^{\text{true}}.$$

Loop corrections are expected to shift  $d_s^{\text{true}}$  into  $[2.0, 2.079]$ , consistent with LQG  $[2.0, 2.2]$  and CDT  $[2.0, 2.5]$  numerical results.

Delporte, Sen and Toriumi (2024) [7] study “Dirac walks” on the Bethe lattice and find that the spectral dimension of the Dirac walk *matches that of the simple random walk* ( $d_s \rightarrow 3$  asymptotically for  $d \geq 2$ ), independently of

edge orientation. This result applies to walks without the EPRL intertwiner degeneracy; it provides an independent confirmation that loop corrections and walk-type choices affect  $d_s$ , and sets a bound that the oriented non-backtracking walk considered here must be distinguished from the Dirac walk of [7]. The two walks are physically distinct: the present walk uses the non-backtracking rule combined with the EPRL  $K_{\text{node}} = 2$  factor; the Dirac walk of [7] uses signed steps without this intertwiner degeneracy.

[One remaining open step] The argument above is rigorous at the level of a single EPRL vertex under the saddle-point approximation. The remaining open step — that iterative coarse-graining preserves  $K_{\text{node}} = 2$  at each scale without cancellation — is identified as Open Problem 1 of TEG vH2 [9]. If confirmed by tensor-network renormalization,  $D_{\text{eff}} = \ln 8$  is complete from first principles. If falsified, the walk would predict a different  $d_s$ , and TEG would require revision.

## 6 Continuum Limit and Falsifiable Scale

In the continuum limit  $a \rightarrow 0$ , the link variables and densitized triads of the tetrahedral lattice map onto Ashtekar variables following the same procedure used in Chrono-Grid Dynamics.

The isoperimetric form  $A(z) \propto z^{2/3}$  introduced in Section 2 is employed solely for the relative comparison that selects  $z_{\text{fund}} = 4$  at fixed vertex density. The overall multiplicative constant appearing in  $A(z)$  is conventional and can be absorbed into the definition of the lattice spacing  $a$ .

The physical value of the effective tension is taken from previous TEG derivations based on holographic roughness and equipartition,

$$\sigma_{\text{eff}} = 0.1088.$$

[Epistemic status of  $r_J$ ] The value  $r_J \approx 0.62 \text{ kpc}$  is obtained from Eq. (??) using  $\sigma_{\text{eff}} = 0.1088$  (derived from the tetrahedral axiom via holographic equipartition, TEG v8 Appendix E) and the critical density  $\rho_0$  as an external input. A fully axiomatic derivation of  $r_J$  in physical units — without  $\rho_0$  as an input — has not been achieved; this is Open Problem 7 of TEG vH2 [9]. A candidate resolution via a UV–IR equilibrium condition is presented in TEG vH2 Appendix L (conjecture, pending tensor-network renormalization verification). The value  $r_J = 0.62 \text{ kpc}$  is therefore presented here as an empirically motivated reference scale (7% agreement with the SPARC empirical median  $r_J^{\text{SPARC}} \approx 0.58 \text{ kpc}$ ), not as a zero-parameter prediction derived solely from the tetrahedral axiom.



This yields the Jeans scale

$$r_J = \sqrt{\frac{\sigma_{\text{eff}}}{4\pi G \rho_0}} \simeq 0.62 \text{ kpc} \left( \frac{\rho_0}{0.1 M_{\odot} \text{pc}^{-3}} \right)^{-1/2}.$$

Both  $r_J$  and the fractal dimension  $\Phi_F = 3 - \ln 8 \approx 0.9206$  constitute falsifiable predictions. If SPARC rotation curves, Euclid void profiles, or laboratory box-counting measurements reject these values, then the selection  $z_{\text{fund}} = 4$  is falsified.

## 7 Discussion: CGD vs TEG

Table 1: Comparison

|                | CGD [5]          | TEG (this work)                                 |
|----------------|------------------|-------------------------------------------------|
| $z$            | postulated 6     | derived 4 by $\max s(z)$                        |
| Cube           | fundamental      | $3! = 6$ composite $T_{\sigma}$                 |
| $\mathbb{R}^4$ | not discussed    | 5-cell, $D_4$ , 8 cells = $2z$                  |
| Number         | $\tau$ undefined | $\Phi_F = 3 - \ln 8$ , $r_J = 0.62 \text{ kpc}$ |
| Falsifiable    | No               | Yes                                             |

## 8 Conclusion

Cubic symmetry is emergent. Tetrahedral  $z_{\text{fund}} = 4$  is fundamental, selected by entropy, generating cube via  $3! = 6$  in  $\mathbb{R}^3$  and hypercube via  $2z_{\text{fund}} = 8$  in  $\mathbb{R}^4$ , with signature  $\ln 8$ .

**Scope.** The results of this paper are: (i) an analytic proof that  $z_{\text{fund}} = 4$  maximises holographic entropy density among physical coordination numbers, independent of the normalisation convention for  $A(z)$  (Theorem 1); (ii) the emergence of cubic symmetry from tetrahedral coordination via  $3! = 6$  (Theorem 2); (iii) the 4D lift to the 5-cell and tesseract with  $2z_{\text{fund}} = 8$  cubic cells (Section 4); and (iv) the spectral dimension  $D_{\text{eff}} = \ln 8$ , derived from the EPRL intertwiner degeneracy  $\dim \mathcal{H}_{\text{int}}^{(4)} = 2$  under the non-backtracking walk on the Bethe lattice (Theorem 3). The derivation of  $r_J$  in physical units (Remark 6). The coarse-graining preservation of  $K_{\text{node}} = 2$  (Open Problem 1 of TEG vH2 [9]) is the single remaining step needed to make  $D_{\text{eff}} = \ln 8$  fully rigorous from first principles.

## A Reproducibility: Entropic Selection Code

The following Python 3 code reproduces Lemma 1,  $s(z)$  table and Theorem 1,  $z_{fund} = 4$ . It is deposited in `/src/entropy.py` of the Zenodo record.

Listing 1: Entropic selection: analytic bound and numerical verification.

```
import numpy as np

# Analytic bound:  $s(z)$  strictly decreasing for  $z \geq 4$ 
z_star = np.e**(1.5) / 2
print(f"z* :=  $e^{(3/2)}/2$  := {z_star:.4f} < 4")
print("=>  $s(z)$  strictly decreasing for  $z$  in {4, 6, 8, 12, 20}")
print()

# Numerical verification (normalization-independent)
z_vals = [4, 6, 8, 12, 20]
s = {z: np.log(2*z) / z**(2/3) for z in z_vals} #  $C$  cancels
winner = max(s, key=s.get)
assert winner == 4, "Entropic selection failed"
print(f"argmax  $s(z)$  := {winner} - VERIFIED")
print(f"Margin  $s(4)/s(6)$  := {s[4]/s[6]:.6f}")

# Normalization independence: same margin for any  $C$ 
for z0 in [4, 6, 8, 12]:
    C = 2 * (1/z0)**(2/3)
    s_norm = {z: np.log(2*z)/(2*C*z**(2/3)) for z in z_vals}
    ratio = s_norm[4]/s_norm[6]
    assert abs(ratio - s[4]/s[6]) < 1e-12
print(f"Normalization independence: - VERIFIED - (ratio invariant)")
```

## References

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- [9] M. Á. Franco León, *Tetrahedral Emergent Gravity vH2: The Minimal Simplex Principle and the Quaternion Projection Paradigm*, Working draft, June 2026. <https://github.com/MiguelAngelFrancoLeon/mfsu-tetraedro> [Open Problem 1 (coarse-graining preservation of  $K_{\text{node}} = 2$ ), Open Problem 7 (derivation of  $r_J$  in physical units), Appendix K (EPRL intertwiner derivation of  $b_{\text{eff}} = 8$ ), Appendix L (UV-IR equilibrium conjecture for  $r_J$ ).]
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